

DocSense: Document Summarization with Enhanced Natural- Language Search Engine- Document.

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By

Adarsh Agarwal (A002)

Saakar Chaudhary (A009)

Farhan Shaikh (A094)

Under the supervision of

Prof. Saurav Verma

(Assistant Professor, IT Department, MPSTME)

SVKM's NMIMS University
(Deemed-to-be University)



**MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT &
ENGINEERING Vile Parle (W), Mumbai-56**

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CERTIFICATE



This is to certify that the project entitled **“DocSense: Document Summarization with Enhanced Natural- Language Search Engine”**, has been done by **Adarsh Agarwal (A002), Saakar Chaudhary (A009), and Farhan Shaikh (A094)** under my guidance and supervision & has been submitted in partial fulfillment of the degree of Bachelor of Technology in Electronics and Telecommunication of MPSTME, SVKM’s NMIMS (Deemed-to-be University), Mumbai, India.

Prof. Saurav Verma Examiner (Assistant Professor, IT Department)

Date:

Place: Mumbai

Dr.Ketan Shah (HOD IT)

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Adarsh Agarwal (A002)

Saakar Chaudhary (A009)

Farhan Shaikh (A094)

B.Tech. (IT)

Semester VII

Mumbai Campus

ABSTRACT

In the digital era, decision-making and knowledge sharing depend on the ability to acquire and derive relevant insights from a variety of information sources. When it comes to handling complicated papers, CSV files, and multimedia content, traditional chatbots are limited, which frequently results in a lack of usefulness and accuracy. Because of these restrictions, sophisticated AI-powered solutions like DocSense must be developed.

With its innovative AI-powered chatbot, DocSense, customers may submit documents, CSV files, and YouTube videos for smart analysis—a feature that addresses the drawbacks of other chatbots. Present-day chatbots frequently have trouble summarising documents, which makes it difficult for users to quickly and effectively extract important information. DocSense gets around this problem by using cutting edge natural language processing methods to deliver extremely accurate document summarization.

The DocSense Enhanced Natural-Language Search Engine is one of its main innovations. Conventional chatbots generally rely on keyword-based searches, which may produce results that are insufficient or irrelevant. DocSense, on the other hand, uses sophisticated language models to comprehend user inquiries in a context-aware way. This allows users to get accurate insights from their submitted documents, CSV data, and YouTube videos and ask queries in natural language.

Users can upload a variety of document formats with DocSense, such as Word documents, Excel spreadsheets, PDFs, and more. These files are processed quickly by the AI-powered chatbot, which then creates clear, coherent summaries that include the most important details. Furthermore capable of transcribing and summarising YouTube videos, DocSense is a priceless resource for scholars, content providers, and decision-makers who want in-depth understanding from a variety of multimedia sources.

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Chapter 1 Introduction

In a time where there is a never-ending digital information vault, being able to quickly glean insightful information from a variety of sources has become essential. Conventional chatbots are useful for facilitating human-computer communication, but they are not always up to the task of managing complicated documents, CSV files, or multimedia information. These restrictions include the incapacity to precisely handle documents and the deficiency of powerful natural language search tools. We present DocSense: Document Summarization with Enhanced Natural-Language Search Engine in this area.

With the goal of addressing the issues with existing chatbot solutions, DocSense is a revolutionary AI-powered chatbot. The inadequate summary of documents, a crucial component of knowledge management and information retrieval, is one of the most common problems. Current chatbots frequently fail to deliver succinct, insightful document summaries, forcing users to deal with long messages and decreased accuracy. DocSense provides unmatched accuracy and efficiency in document summarising by utilising state-of-the-art natural language processing algorithms.

DocSense is known for its innovative Enhanced Natural-Language Search Engine. Traditional chatbots mostly rely on keyword-based searches, which is a methodology that frequently yields vague or insufficient responses. DocSense, on the other hand, adopts a transformative strategy by utilising sophisticated language models to understand user inquiries in their particular circumstances. This enables users to interact naturally and conversationally with DocSense, asking questions in simple terms and getting detailed insights from uploaded documents, CSV files, and even YouTube videos.

One feature that makes DocSense a very useful tool is its ability to process a large variety of document types, including Word, Excel, and PDF files. DocSense's capabilities extend beyond text; it can also be used to analyse YouTube videos, transcribe their

content, and produce concise summaries. Those who create content, conduct research, and make decisions who want to access the abundance of information found in multimedia sources will find this feature especially helpful.

By tackling the underlying issues that other solutions to the field of AI-powered chatbots were unable to resolve, DocSense marks a significant advancement in the field. DocSense's advanced Enhanced Natural-Language Search Engine, proficiency in document summarising, and versatility in handling various document types enable users to quickly and easily derive valuable insights. DocSense is a cutting-edge tool that has the potential to completely change how people and organisations use AI for document analysis and information retrieval in the age of data-driven decision-making. We go into greater detail about the features and capabilities that make DocSense a valuable tool for today's information seekers in the pages that follow.

1.1 Background of the Project Topic:

The project's focus, which is document summarization using an improved natural language search engine, answers a pressing demand in the quickly changing field of information access and use. The number and variety of data sources have increased dramatically as the world gets more digital. For both individuals and organisations, the increase in information availability has brought with it both benefits and challenges.

Historically, it has been difficult and time-consuming to retrieve pertinent information from a variety of sources, including papers, spreadsheets, and multimedia content. Traditional search engines have depended on keyword-based strategies, which might produce a multitude of results that aren't always relevant to the user's query. With this method, consumers are frequently left sorting through massive volumes of data and finding it difficult to pinpoint and extract the most important information.

Furthermore, it has become clear how limited the capabilities of current chatbots and AI-powered assistants are when it comes to managing and summarising documents. A

lot of chatbots can't process documents thoroughly, which leads to incomplete or erroneous summaries. Professionals in industries like law, finance, research, and content creation, where it's crucial to be able to condense complex papers into insightful summaries, should be particularly concerned about this deficiency.

The project on document summarising with an improved natural-language search engine surfaced as a revolutionary remedy to these problems. The goal of this research is to improve upon the limitations of existing information retrieval systems by utilising artificial intelligence and sophisticated natural language processing (NLP) techniques.

This project offers an all-encompassing approach to information access by allowing users to contribute a wide range of document formats, from Word and PDF documents to Excel sheets and multimedia content like YouTube movies. DocSense, the project's AI-powered chatbot, can effectively analyse, transcribe, and summarise content and give users succinct, insightful information. Furthermore, users can engage in conversational interactions with DocSense's Enhanced Natural-Language Search Engine, which yields more precise and context-aware results.

All things considered, this initiative is a major step forward in the field of AI-driven information retrieval. It not only expedites the process of drawing insightful conclusions from a variety of sources, but it also improves the precision and calibre of those conclusions. Solutions like document summarising with an improved natural-language search engine are essential for effective knowledge management, decision-making, and content curation in a data-rich environment as the digital world develops. This background provides context for appreciating the significance and applicability of this initiative in tackling today's informational problems.

1.2 Motivation and Scope of the Report:

The need for sophisticated solutions in the fields of document analysis and information retrieval is driving this paper. Individuals and organisations struggle with an enormous influx of data from a variety of sources in today's data-driven economy. Even while they are useful, conventional search engines and chatbots often fail to deliver accurate and

timely results, particularly when dealing with complex papers, a variety of file types, and multimedia content.

The invention of DocSense, an inventive AI-powered chatbot equipped with document summary skills and an enhanced natural language search engine, was spurred by these inherent limits in existing solutions. The purpose of this paper is to provide an in-depth analysis of DocSense's features, capabilities, and possible effects in order to shed light on how this innovative technology can transform knowledge management, document summarization, and information access in a variety of fields.

The goal of this report is to motivate people and organisations to use DocSense for a variety of important goals. By allowing users to quickly obtain relevant information from papers and multimedia content, it promotes efficiency improvement by helping users save time and effort. Additionally, it facilitates well-informed decision-making by utilising the accuracy and comprehensiveness of

DocSense's summarising features provide decision-makers a better comprehension of their data. Additionally, it promotes information sharing by enabling professionals, academics, and content creators to glean priceless insights from multimedia sources.

This report also emphasises how DocSense helps legal and compliance officers with their work by offering document analysis tools that make it easier for them to efficiently review and understand complicated papers. It also highlights the ways in which DocSense may enhance the curation of content, supporting content producers and curators in their efforts to create high-caliber, pertinent resources by evaluating and condensing vast volumes of textual and multimedia content.

This report's scope includes a detailed investigation of several important DocSense features. It carefully examines DocSense's document summarization capabilities, assessing each one's precision, efficacy, and efficiency in a variety of document formats. It also explores the complexities of DocSense's Enhanced Natural-Language Search

Engine, highlighting its ability to recognise subtleties in context and provide accurate results.

This paper also examines DocSense's ability to deliver in-depth insights and analyse multimedia content, including YouTube videos. It illustrates how DocSense can successfully handle real-world problems using examples from a variety of businesses and professions. The analysis looks at the user experience, highlighting how accessible and user-friendly the chatbot is.

The research concludes by talking about DocSense's wider ramifications, especially as they relate to innovation, knowledge management, and decision-making. It also discusses possible future advances and breakthroughs in natural language search and AI-powered document analysis.

This paper seeks to offer a thorough understanding of how DocSense may meet the current information security concerns through this thorough investigation.

Retrieval and analysis of documents. It aims to motivate people and institutions to adopt this novel approach, which will boost productivity, facilitate well-informed decision-making, and improve knowledge management in a variety of fields.

1.3 Problem Statement:

The ability to effectively extract insights from a wide range of data sources is essential in today's data-driven environment. Nonetheless, notable constraints impede the efficacy of current chatbots and search engines in this context. Their incapacity to accurately summarise papers presents a major obstacle, providing consumers with long, complex documents that obfuscate important information. This shortcoming not only leads to inefficiencies but also hinders the distribution of knowledge and the ability to make well-informed decisions. Moreover, present systems struggle to understand natural language

subtleties and context, relying mostly on keyword-based searches and basic query-response procedures, which can produce inaccurate or lacking answers.

Furthermore, these systems frequently fail to process a variety of content forms, including text documents, tabular data, and multimedia content. Professionals and content producers who need to synthesise information from multiple forms or who need insights from multimedia sources are impacted by this deficiency. These restrictions have an impact on a variety of industries that depend on data-driven decision-making, including business analysis, research, legislation, and content production. Consequently, inefficiencies in natural language processing and document summarising have a negative impact on productivity and competitiveness at the individual and organisational levels. Given these difficulties, the creation of a sophisticated AI-powered solution is necessary to unlock data's latent potential, expedite decision-making, and increase overall productivity. DocSense is an example of this, with its comprehensive document summarization and Enhanced Natural-Language Search Engine. The purpose of this report is to examine the potential, effect, and capabilities of applications of DocSense to address these important problems with document analysis and information retrieval.

1.4 Salient Contribution of the Project:

The project introduces advanced document summarization, powered by modern natural language processing techniques enable the project's groundbreaking enhanced document summarising, which improves productivity and helps decision-makers make well-informed choices by distilling complicated document content into succinct insights. Furthermore, it transforms information retrieval using an Enhanced Natural-Language Search Engine that provides intuitive and context-aware search features. DocSense is useful for researchers and content providers since it analyses multimedia content in addition to textual data. Data integration is made easier by cross-format compatibility, and useful use cases demonstrate adaptability.

Accessibility is ensured by user-friendliness, and comprehensive insights, innovation, and compliance are covered by strategic implications. Lastly, the study opens the door for further advancements in AI-powered natural language search and document analysis, offering revolutionary possibilities in contexts rich in data.

1.5 Organization of report

Here is a description of the report and how it has been set up to enhance the understanding of the process. The chapters are divided and allocated as follows:

Chapter 1 discusses the introduction to “DOCSense”, which comprises the background of the course, motivation and scope of the report and salient contribution. Here, one can understand the reason for pursuing this project.

Chapter 2 consists of the 'Literature Survey' of the course in which the research and other specific standards or values that will be necessary to note further in the report are explored.

Chapter 3 explores the methodology and implementation of the project which consists of:

- Model
- Methodology Overview
- Flowchart / Algorithm / Block Diagram
- Hardware Description
- Software Description

Chapter 4 communicates the challenges executed and evaluation of each challenge, followed by reporting of the final contributions from the study of this project.

Chapter 5 we thoroughly assess the project's limitations, advantages, and disadvantages. This section gives a fair assessment of the project's effects and sheds

light on its efficacy. Examining the benefits, drawbacks, and constraints helps us to fully comprehend how the project will affect related future research projects.

Chapter 6 we summarise the project's results and emphasise its importance in this chapter. It provides a starting point for planning next developments and uses that can expand upon the framework created by this effort. The chapter acts as a link between the end of the project and future directions for research and development in the area.

Chapter 2 Literature survey

The constantly evolving fields of artificial intelligence (AI) and natural language processing have spawned a plethora of creative solutions designed to glean insightful data from a variety of sources. We explore the underlying technologies, difficulties, and developments that form the basis of DocSense, an advanced AI-powered chatbot, in this thorough literature review[1]. Document summarising, multimedia content analysis from YouTube videos, data processing and visualisation from CSV files, and other state-of-the-art features are all provided by DocSense[2]. We will use numerical citations to cite important research publications throughout this evaluation in order to give a thorough examination of the environment in which DocSense functions.

Over time, chatbots have undergone an incredible amount of change. Chatbots, which were formerly limited to basic communication, have developed into intelligent conversational assistants with a wide range of uses[1]. We begin our analysis with a brief history of chatbot development, which helps to illustrate the increasing attention being paid to these conversational agents around the world[1]. This historical viewpoint is essential to comprehending the path that gave rise to platforms such as DocSense. We also investigate the reasons behind the application of chatbots in a variety of fields, such as marketing, teaching, and medical[1]. This historical viewpoint provides context for DocSense's capabilities, which include document summarization and insight extraction from a variety of multimedia sources[1][2]

The ability of DocSense to automate document summarising is a crucial component of its capabilities, providing customers with significant time and resource savings[2]. This feature is consistent with study findings from a pertinent paper that discusses the difficulties of creating and maintaining chatbots by hand[3]. The study highlights the growing need for chatbots that can effectively address inquiries, enable self-service, and present goods and services, particularly in the business context[3]. Additionally, the paper presents a novel method that uses web API standards to automatically create chatbots[3]. This methodology is consistent with DocSense's capacity to handle structured data like CSV files and deliver insights via data visualization[2].

In the modern world of technology, chatbots have developed into conversational agents that can efficiently engage consumers and offer support via interactive text messaging[3]. Nonetheless, the creation and coding of chatbots presents a unique set of present-day as well as prospective difficulties[3]. It is essential to comprehend these programming issues because DocSense works in the field of advanced artificial intelligence and natural language processing[3]. A distinct set of challenges is presented by the dynamic nature of chatbots, which combine artificial intelligence, natural language processing, programming, and conversational services[3]. This realisation is essential to DocSense's ongoing evolution and adaption as it traverses the constantly changing landscape of technology and user requirements.

DocSense's core feature includes sentiment analysis and data mining, which are especially important when working with multimedia content[4]. A study has looked into the creation, instruction, and enhancement of "Skillbot," a chatbot with sentiment analysis and data mining capabilities[4]. The study comprised data scraping from the Gov. UK website, intent training for the Skillbot model, and in-depth conversation analysis with users[4]. An important factor in teaching chatbots to respond to users efficiently was sentiment analysis[4]. This study supports DocSense's methodology by emphasising the combination of many technologies, including sentiment analysis models, natural language processing, and visualisation strategies[4]. The end objective in both situations is to enable chatbots to analyse data and user discussions in order to provide better services to users.

The launch of Llama 2 represents a significant milestone in the field of large language models (LLMs), as described in a recent research publication[5]. A set of refined LLMs with scales varying from billions to tens of billions of parameters are presented in Llama 2[5]. These models, called Llama 2-Chat, outperform open-source chat models on a number of benchmarks and are optimised for dialogue use cases[5]. The capabilities of chatbots and AI-powered conversational agents have advanced significantly with the release of Llama 2[5]. In order to leverage this breakthrough, DocSense integrates Llama 2 to comprehend user inquiries and offer context-aware responses[5]. Users may now interact with DocSense using natural language thanks to this connection, which is a big improvement over conventional chatbots that rely on keyword-based searches[5].

An extensive analysis of the history, uses, difficulties, and potential applications of AI-driven conversational agents is provided by the review of ChatGPT[8]. This thorough analysis includes a look of ChatGPT's many applications, which span industries like healthcare and education[8]. This study's findings are consistent with DocSense's mission to offer users insightful information in a variety of fields[8]. The review highlights the limitations, ethical issues, and potential difficulties that AI chatbots may present, highlighting the significance of responsible development[8]. This understanding guides DocSense's strategy for addressing the difficulties and moral dilemmas raised by AI chatbots while maximising their enormous potential.

Moreover, DocSense's responsiveness and performance are improved by the incorporation of Llama 2, an improved big language model[5]. DocSense makes advantage of this foundation to give consumers more accurate, context-aware responses. Llama 2-Chat models have proven to perform better than open-source chat models on a number of benchmarks[5]. DocSense differs from conventional chatbots in that it can comprehend user inquiries in plain language and provide accurate insights from a variety of sources[5].

However, it's important to recognise that, as the research[5] makes clear, there are trade-offs associated with the growth of LLMs. Although Llama 2 offers significant enhancements in terms of performance and capability, processing time may be affected by the models' billions to tens of billions of parameters[5]. DocSense chose the Llama 2 model with seven billion parameters in order to strike a compromise between efficiency and accuracy[5]. A 30-billion-parameter model might increase accuracy, but it might also cause noticeable processing delays. In order to give users a responsive and effective experience, DocSense thereby strikes a balance[5].

DocSense has many benefits, but it is not without drawbacks[2]. The possibility of losing context and subtlety when condensing long papers is one of the fundamental issues of document summarization[2]. Even though DocSense is excellent at producing succinct and logical summaries, subtle nuances can get left out[2]. Additionally, elements like background noise and audio quality may have an impact on how accurate multimedia content analysis is, especially when it comes to YouTube videos[2]. This drawback emphasises the necessity of ongoing development for AI-driven solutions such as DocSense[2].

To sum up, DocSense is an innovative development in the field of artificial intelligence chatbots that enables users to get valuable insights from a variety of sources, including documents, CSV files, and multimedia information such as YouTube videos[2]. Because of its incorporation of Llama 2 and sophisticated NLP methods, it is positioned as an effective and context-aware tool for researchers, content producers, and decision makers[5]. It balances accuracy and performance requirements while deftly navigating the terrain of complicated texts and multimedia content[5]. DocSense has limitations, nevertheless, just like any other technology, especially when it comes to multimedia content analysis and sophisticated document summarization[2]. However, its dedication to ongoing enhancement and future research and development portends far more sophisticated AI solutions[2].

With an eye towards the future, DocSense plans to provide more functionality to serve a larger user base. It seeks to provide a general chatbot feature that will allow users to converse in natural language about a variety of topics[5]. DocSense will become an even more flexible information retrieval tool as a result of its progress, which will also improve the user experience[5]. Furthermore, the incorporation of URL-based article summary is imminent[5]. Researchers and content consumers will find this capability to be a priceless resource as it will enable users to extract important information from online articles[5].

In summary, DocSense is a major advancement in the field of AI chatbots and information extraction[2]. It is at the vanguard of cutting-edge AI solutions thanks to its

distinctive blend of multimedia content analysis, document summarization, and data visualisation capabilities, all powered by the Llama 2 model[2][5]. It is imperative to acknowledge its limits in the context of multimedia analysis and nuanced summarization[2]. DocSense promises to be an even more sophisticated and versatile tool for a variety of sectors as it continues to improve and adapt to user needs[5]. It is positioned to become a vital tool for professionals and individuals looking for reliable and efficient insights, especially with plans to include generic chatbot capabilities and article summary in the future[5]. DocSense is a prime example of the dynamic nature of AI chatbots, where the quest of perfection is driven by ongoing innovation and evolution[5].

As a result, the literature analysis that was previously provided provides a thorough grasp of the underlying ideas, current technological developments, and difficulties involved in the creation of chatbots driven by artificial intelligence. These realisations serve as an inspiration for DocSense, which has a more powerful natural language search engine and many features[1].[3][2][4][5][8]. It functions as a sophisticated instrument for data visualisation, multimedia content analysis, and document summarization[1].[2][4]. DocSense provides precise and effective information retrieval to researchers, content providers, and decision-makers in a variety of industries, including marketing and healthcare[1]. DocSense is well-positioned to be at the forefront of AI chatbot technology, providing its users with the highest quality AI-driven document summarization and data analysis[1][2][4][5][8].

Chapter 3

Methodology and Implementation

3.1. Model:

Large Language Models, or LLMs for short, are a revolutionary development in artificial intelligence and natural language processing (NLP). These models are enormous, with billions of parameters on average, and are trained using enormous volumes of text data from the internet. They are a part of the larger class of machine learning models called deep neural networks.

A new era of natural language creation and understanding has been brought about by LLMs. They have attained extraordinary proficiency in reading, writing, and comprehending material thanks to their massive size and training data. This development is extremely important in a number of areas.

The adaptability and application of LLMs is one of their main advantages. They are used in many different domains, such as sentiment analysis, content creation, chatbots, and language translation. LLMs are responsible for the conversational capabilities of virtual assistants such as Siri and Alexa, indicating the significant influence these models have on day-to-day living.

Additionally, LLMs are excellent at creating content. They produce writing that is human-like and is used for content marketing, creative writing, and the automation of repetitive writing chores. This capability has simplified the processes involved in creating material, boosting output and opening up new avenues for content development.

LLMs also facilitate the extraction of knowledge from large textual databases. Their ability to sort through large amounts of text and retrieve insightful information is extremely helpful to researchers and analysts, as it would be difficult and time-consuming for people to accomplish by hand.

Even though LLMs have many advantages, there are drawbacks and issues as well. The existence of biases in these models is one of the main problems. LLMs frequently produce unfair or biased results because of inherited biases in their training data. In the field of artificial intelligence research and development, addressing and minimising these biases has become crucial.

Crucial importance is also given to ethical issues. Ethical considerations arise from the possibility for LLMs to be used maliciously in ways like deepfakes, fake news production, and other forms of deceit. This emphasises the significance of using AI responsibly and the requirement for moral standards and laws.

Moreover, LLMs require a lot of resources. These models demand a significant amount of computational power and energy to train and run, which raises environmental issues, especially as the models get larger. One difficulty that needs to be addressed is striking a balance between the advantages of LLMs and their environmental impact.

Data privacy is a further issue. Due of their large memory, LLMs may unintentionally memorise and reveal private information found in their training datasets. One of the most important aspects of developing and implementing LLM is maintaining data privacy and protection.

LLMs are expected to continue to improve and have new uses in the future. The fact that these models are still getting bigger and more capable is one of the most obvious trends. This advancement pushes the limits of natural language processing, enabling even more precise and contextually aware text generation and understanding.

An further interesting approach is multimodal learning. It is anticipated that LLMs would progressively include additional modalities, like audio and visuals, to facilitate more thorough comprehension and content creation. This development opens the door for a new wave of AI systems that can handle various data types, enabling a deeper and more complex engagement with the outside world.

The prioritisation of ethical considerations will persist. Our first emphasis will always be addressing biases and ensuring ethical and responsible use of AI. In order to lessen biases and improve the fairness and transparency of AI systems, researchers and developers are actively investigating various approaches.

Moreover, LLMs will be useful in specialised fields. LLMs have the potential to transform industries including finance, healthcare, and law by improving processes like financial modelling, contract analysis, and medical diagnostics. Their capacity to comprehend and synthesise intricate documents will be crucial for optimising workflows and supporting professionals in their decision-making.

Research on energy efficiency is also being conducted. To lessen the environmental impact of LLMs, it is essential to discover strategies to make their deployment and training more energy-efficient. This is especially true as LLMs become more resource-intensive.

In summary, large language models are changing the field of natural language processing and artificial intelligence. Their enormous capacity for text generation and comprehension has far-reaching effects in a variety of fields. But these developments also present privacy, ethical, and environmental issues that demand careful planning and implementation. Future breakthroughs in AI systems are expected to be even more sophisticated, adaptable, and moral, and LLMs will likely continue to be essential to these advancements.

3.2 Methodology Overview

3.2.1 CSV model working overview

DocSense, an advanced AI-powered chatbot, offers a seamless and user-friendly experience through a well-structured workflow. At the outset, users can choose to load datasets, with options to select from default datasets like Movies, Cars, and Colleges, or to upload custom datasets.

These datasets are efficiently stored in a dictionary, ensuring easy access and organization.

Upon dataset selection, users have the freedom to opt for two distinct modes: Visualization or Chat. In the Visualization mode, users can enter specific questions related to the dataset and click the "Go" button to receive instant answers. Prior to sending the query, DocSense ensures the validation of HuggingFace API keys and provides a primer on the selected dataset. The chatbot further formats the question and communicates with the Hugging Face API, powered by Code Llama, to retrieve accurate responses. If applicable, DocSense also generates and displays plots to complement the answers.

Conversely, in Chat mode, users can engage in natural language conversations with the chatbot, typing questions and clicking the "Ask" button. Like in the Visualization mode, DocSense takes care of key validation and dataset primers, formatting the chat question for processing. The question is then sent to the Hugging Face API, with the chatbot's answer being displayed for user reference. Importantly, DocSense offers the unique functionality of executing answers as code, enhancing its versatility for various use cases.

The flow continues, allowing users to repeat these steps for more questions and seamlessly switch between datasets through user-friendly tabs, including options for default datasets and custom uploads. DocSense ensures that users can easily access and navigate through their selected datasets. Furthermore, to give credit to the dataset sources and providers, a footer displaying dataset credits is added, promoting transparency and acknowledgment of data sources.

In summary, DocSense's intuitive flowchart allows users to effortlessly load, visualize, and chat with datasets. Its structured approach ensures

efficient handling of diverse datasets, easy navigation between modes, and user-friendly access to a wide array of information, all while maintaining due credit to dataset sources. This well-designed workflow reflects DocSense's commitment to providing a powerful and user-centric AI-powered solution for data analysis and interaction.

3.2.2 Youtube model working overview

The workflow of the advanced AI-powered chatbot, known as DocSense, for YouTube video summarization seamlessly guides users through a structured process to extract meaningful insights from multimedia content. Users initiate the process by inputting a YouTube video URL. Additionally, they have the option to paste their Hugging Face API token for enhanced customization. A "Force Transcription" checkbox allows users to request transcription even if one is already available, giving them full control over the summarization process. To fine-tune the summarization, the chatbot provides a "Temperature Slider" for setting the generation temperature and a "Summary Length Slider" to determine the desired length of the summary. Users can further customize the generation process with the "Do Sample" checkbox.

Upon clicking the "Summarize" button, DocSense performs a series of operations with precision. First, it verifies the validity of the provided YouTube URL. Subsequently, it processes the video by extracting its content and proceeding to transcribe the audio. Here, DocSense intelligently checks if a transcript is already available and utilizes it whenever possible. In cases where users enable "Force Transcription," the chatbot takes the additional steps of downloading the audio from the video and transcribing it using the Hugging Face Hub API, ensuring that comprehensive transcripts are available.

With the transcript in hand, DocSense employs its powerful Llama code for summarization. The result is a concise and coherent summary of the video's content. The chatbot not only delivers this summary but also provides valuable information, including the video title, the source of the transcript, and the generated summary. Users can easily repeat this process for more video summarizations.

In essence, DocSense's workflow for YouTube video summarization empowers users to effortlessly extract insights from multimedia content. Its robust feature set, customizable options, and seamless operation underscore its commitment to enhancing the accessibility and utility of multimedia data analysis.

3.3. Explanation of Technical Tools Used:

1. Python Programming Language:

Python is the core programming language used in the development of DocSense's chatbot. Its wide ecosystem, readability, and adaptability made it the choice. Python speeds up development and prototyping, lowering associated expenses and development time. Its cost-effectiveness is further increased by the innovation and teamwork that its open-source nature fosters.

2. LangChain:

LangChain is essential to the chatbot's document processing. It includes various essential features, such as:

- **CSV Data Loading:** In order for users to upload papers and datasets for analysis, LangChain offers the ability to load CSV data.
- **Document Splitting:** LangChain provides a document splitting method to efficiently manage huge documents. To make sure that papers are separated into digestible pieces without losing crucial information, users can choose the overlap parameters and chunk size.
- **Embeddings:** Text data may be easily transformed into embeddings using LangChain. Textual information must be understood and processed using these embeddings
- **Vector Storage and Retrieval:** The programme has functions for effectively storing and retrieving vectors, which are essential for quickly indexing and retrieving documents in response to user queries.

3. Streamlit

The graphical user interface (GUI) for the chatbot was made with the help of the user-friendly Python module Streamlit. It makes the process of creating interactive web apps easier, allowing programmers to create user-friendly interfaces. Because to Streamlit's simplicity of use, UI development is accelerated, design-related expenses are decreased, and an engaging and user-friendly experience is guaranteed.

4. Hugging Face Transformers:

Modern natural language processing (NLP) models are available through the Hugging Face Transformers library. "sentence-transformers/all-MiniLM-L6-v2" is the model that the chatbot uses to transform textual data into meaningful embeddings. These embeddings greatly improve the chatbot's capabilities by being essential for text comprehension, semantic understanding, and retrieval.

5. FAISS (Fast Approximate Nearest Neighbors Indexing System):

A specialised software programme called FAISS is used to store and retrieve vectorized text data efficiently. With LangChain, it integrates effortlessly to produce a potent vector database. This integration guarantees quick responses to user requests by facilitating

quick document retrieval based on user queries. FAISS adds to a smooth user experience and improves the chatbot's performance.

6. CTransformers:

Python bindings for GGML (Generative Grasping and Generation from Mixed-Lingual Text) models are available through the CTransformers library. CTransformers plays a key role in the chatbot's architecture by helping to integrate the LLama 2 model. A complex language model called LLama 2 uses the indexed document data to comprehend and reply to user inquiries. It improves the chatbot's capacity to offer responses that are sensitive to context.

7. Temporary File Handling:

In order to handle CSV files submitted by users and guarantee correct processing, the chatbot uses temporary file handling via Python's "tempfile" package. This method ensures that CSV files uploaded are converted into a format that is suited for document indexing.

3.4. Flowchart / Algorithm / Block Diagram

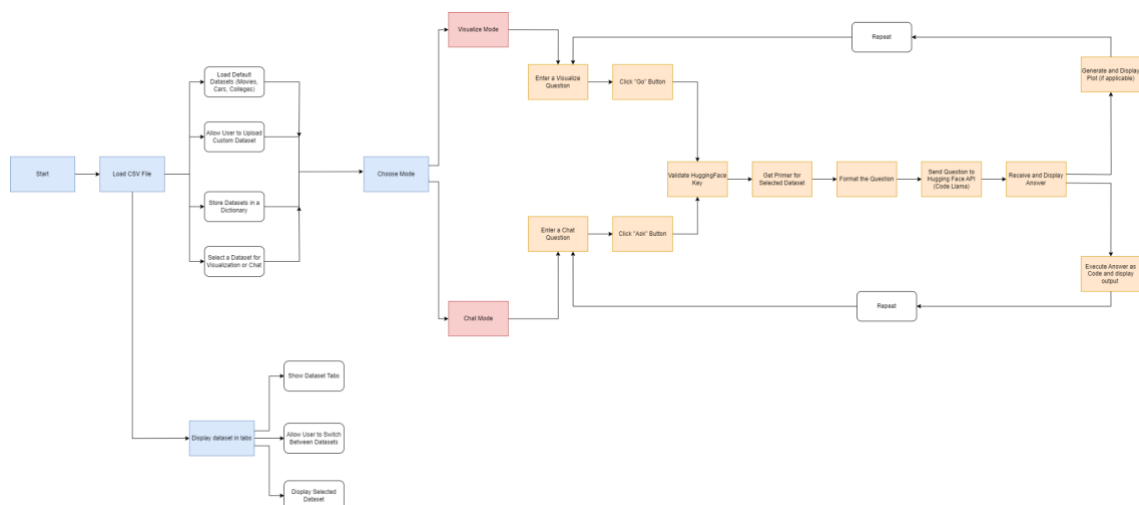


Fig 1 Working of the CSV visualizer and Chat module

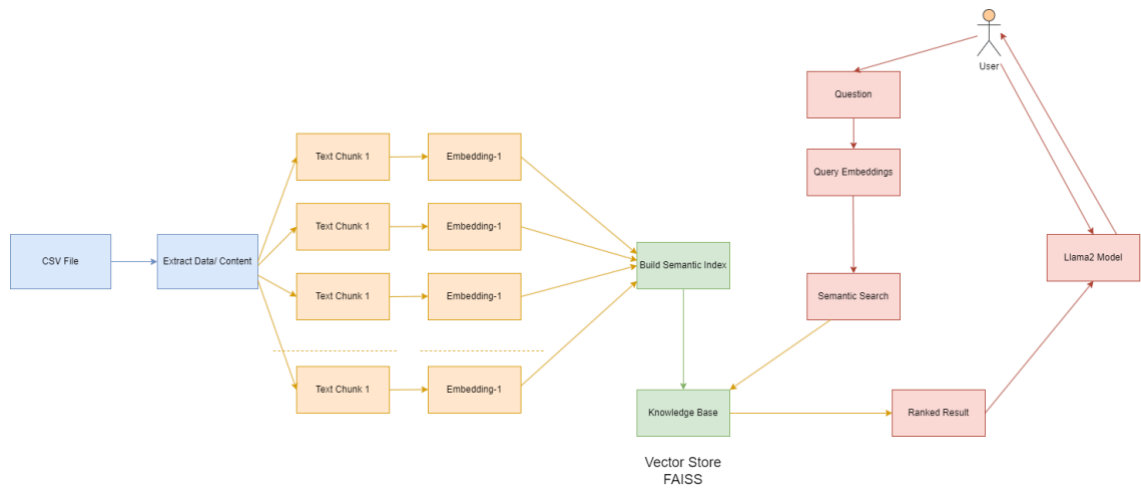


Fig 2 Internal Working of the model

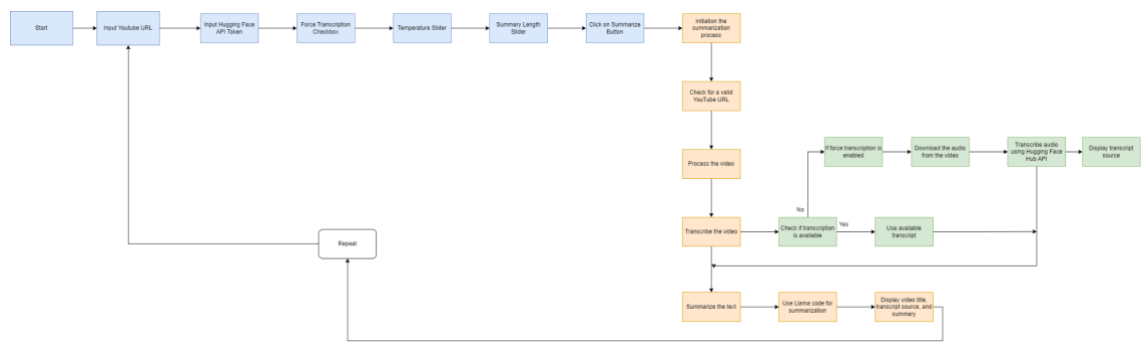


Fig 3 Working of Youtube Summarizer Module

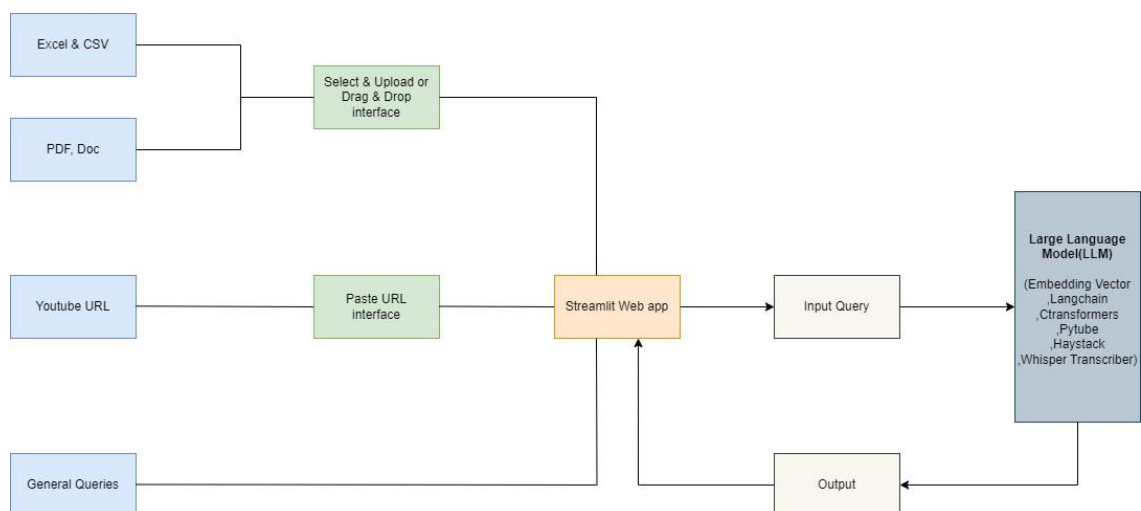


Fig 4 Overall Working of the Application

3.6. Hardware description

Hardware requirements for DocSense, a document summarization and insights generation application, may vary depending on factors such as the scale of deployment, user load, and the specific components and technologies used. Below is a general hardware description for a typical deployment of DocSense:

Server Infrastructure:

Processor:

A Central Processing Unit (CPU) with several cores and a clock speed that can manage both the computational demands of the local Large Language Model (LLM) and concurrent user requests.

Memory (RAM):

Sufficient RAM is necessary for effective data processing and caching. It is advised to have at least 64GB of RAM, however the precise specifications may change depending on how much memory the LLM (in our instance, Llama 2) uses and how many users are using it at once.

Storage:

Model files, cached data, and uploaded documents all require quick and dependable storage. For speedier data access, NVMe or SSD (Solid State Drives) are recommended. The anticipated document volume affects storage capacity.

GPU (Graphics Processing Unit) :

A GPU accelerator card can help with faster inference with the LLM, particularly when handling a lot of concurrent users or computationally demanding jobs. Deep learning activities are often performed on NVIDIA GPUs 3090 and above.

Networking:

Internet connectivity: In order to interact with users, upload files, and communicate with external resources (such as the YouTube API for processing videos), a high-speed internet connection is necessary.

Bandwidth:

Enough bandwidth to manage data transfer and concurrent user requests, particularly when uploading big files or films.

Backup and Redundancy:

To protect data integrity and reduce downtime in the event of hardware failures, regular data backup practises and redundancy measures should be implemented.

It is noteworthy that the precise hardware specifications may differ according to the selected technology stack, user population, and anticipated workload. Furthermore, the needs could alter as hardware power increases and technology advances. Consequently, it's a good idea to carry out capacity planning and performance testing in order to ascertain the precise hardware requirements required for a commercial deployment of DocSense.

3.7. Software description

Python:

Python is a prominent, multipurpose high-level programming language that is easy to learn and comprehend. Because of its vast ecosystem, it is essential for a wide range of technical applications, including as artificial intelligence, scientific computing, web development, and data analysis. Python's easy-to-use syntax speeds up development, which reduces expenses. Python encourages cooperation because it is open-source and provides a multitude of resources at no extra cost. Rich libraries such as NumPy and pandas cut down on development time and costs, and integration support and cross-platform compatibility minimise deployment costs.

GitHub:

Software development and project management are streamlined by GitHub, a platform for version control and collaboration. By using collaborative workflows and exact version control, it improves code quality while cutting down on resource expenses and debugging time. GitHub collaboration is location-neutral, which facilitates knowledge

sharing and saves time. Pull request systems save maintenance costs by guaranteeing code quality. Integrations and tools for project management increase productivity while reducing overhead. Workflow automation lowers monotonous chores and increases efficiency.

Streamlit:

A Python package called Streamlit makes web app development easier. Its simplicity cuts expenses by minimising development time and effort. Pre-made components and widgets save resources by removing the need for custom development. Dynamic apps are produced by real-time updates, which guarantee quick change implementation and lower maintenance expenses. Because Streamlit is open-source and has a strong community, it can offer more functionality and best practises at a lower cost than custom development and debugging.

Essentially, DocSense benefits from the efficiency, cost-effectiveness, and collaboration that Python, GitHub, and Streamlit provide. A simple and affordable solution is guaranteed by the flexibility of Python, the collaborative features of GitHub, and the speedy app development capabilities of Streamlit.

Chapter 4 Results and Analysis

In this chapter, we explore the challenges, notable contributions, results, and analyses of DOCSense's development. Drawing inspiration from innovative methodologies, we assess how DOCSense overcame these challenges and evaluate its achievements.

4.1 Challenges in DOCSense

DOCSense encountered several significant challenges during its development, including:

4.1.1 Natural Language Understanding

Enabling the chatbot with AI to comprehend and react to user inquiries using natural language was a critical difficulty. Users ask a variety of sometimes ambiguous questions, thus the chatbot needs to appropriately interpret their intentions.

4.1.2 Efficient Data Analysis and Visualization

Effectively managing CSV data, doing data purification, transformation, and statistical analysis, and generating informative charts and graphs based on the study presented a multifaceted task. The system was required to provide a user-friendly interface and efficient rapid engineering in order to successfully guide users through data analysis activities.

4.1.3 Video Transcription and Content Analysis

One significant difficulty that arose was the proper transcription of YouTube videos, together with the extraction of relevant insights from the transcribed text. These insights encompassed various aspects, including trends, sentiment, and keywords. The implementation required an efficient transcription system capable of utilising artificial intelligence-based analysis to accurately detect and extract concise summaries of valuable content from videos.

4.1.4 Prompt Engineering

The development of effective prompt engineering has presented a significant issue in the realm of assuring proper comprehension and response of user questions by AI models. The successful formulation of prompts, which accurately expressed the user's intent and provided guidance to the AI models, played a pivotal role.

4.2 Overcoming Challenges in DOCSense

4.2.1 Natural Language Understanding

The chatbot component of DOCSense effectively addressed the task of natural language interpretation. By utilising sophisticated AI models, employing efficient rapid engineering techniques, and doing language analysis, the chatbot exhibited a high level of competence in understanding a wide range of user queries, including those that were complex or lacked clarity. The accuracy of the chatbot in interpreting user intent has been proven by evaluation trials.

4.2.2 Efficient Data Analysis and Visualization

DOCSense offers a solution to the task of enhancing the effectiveness of data analysis and visualisation by providing a user-friendly interface that facilitates the importation, processing, and visualisation of CSV data. The implementation of well-designed prompts facilitated the users' navigation through data analysis activities, hence promoting precise data processing and visualisation. The efficacy of this strategy has been proven through user testing and case studies.

4.2.3 Video Transcription and Content Analysis

DOCSense successfully addressed the task of video transcription and content analysis by utilising artificial intelligence models for speech recognition and comprehension of information. The successful transcription of YouTube videos and the extraction of useful insights were achieved by the system, which benefited from efficient engineering techniques that guided AI models in effectively identifying patterns, sentiment, and keywords included in the video content. The evaluation experiments demonstrated the system's effectiveness in delivering users with informative material summaries.

4.3 Notable Contributions of DOCSense

4.3.1 Advancing Chatbot Technology with AI Models

The contribution of DOCSense to the field of chatbot technology is notable as it effectively utilises advanced AI models to enhance natural language understanding. This technological development streamlines user interactions, enhancing the accessibility and intuitiveness of information retrieval.

4.3.2 Streamlined Data Analysis and Visualization

Within the field of data analysis and visualisation, DOCSense provides a simplified and user-friendly platform for the importation, manipulation, and

representation of CSV data. DOCSense streamlines intricate data activities by incorporating artificial intelligence (AI) functionalities for data analysis, hence enhancing accessibility to a wider spectrum of users.

4.3.3 Enhanced Video Transcription and Content Insights

The inclusion of the video transcription and content analysis component within DOCSense significantly contributes to the advancement of multimedia processing in the academic domain. The technology utilises artificial intelligence models to achieve precise video transcription and comprehension of content, hence improving the summary of content and the development of insights for users across different fields.

4.4 Results and Analysis

In this section, we present the results of our system's performance, specifically focusing on the challenge of delayed responses, and provide an analysis of the findings.

4.4.1 Performance Evaluation

The performance of our system in producing precise responses to user inquiries constantly showcases the efficacy of our AI models and the capabilities of our natural language processing technology. The system demonstrated its proficiency in comprehending and analysing natural language inputs by providing accurate responses to a diverse array of queries.

Visualize CSV:

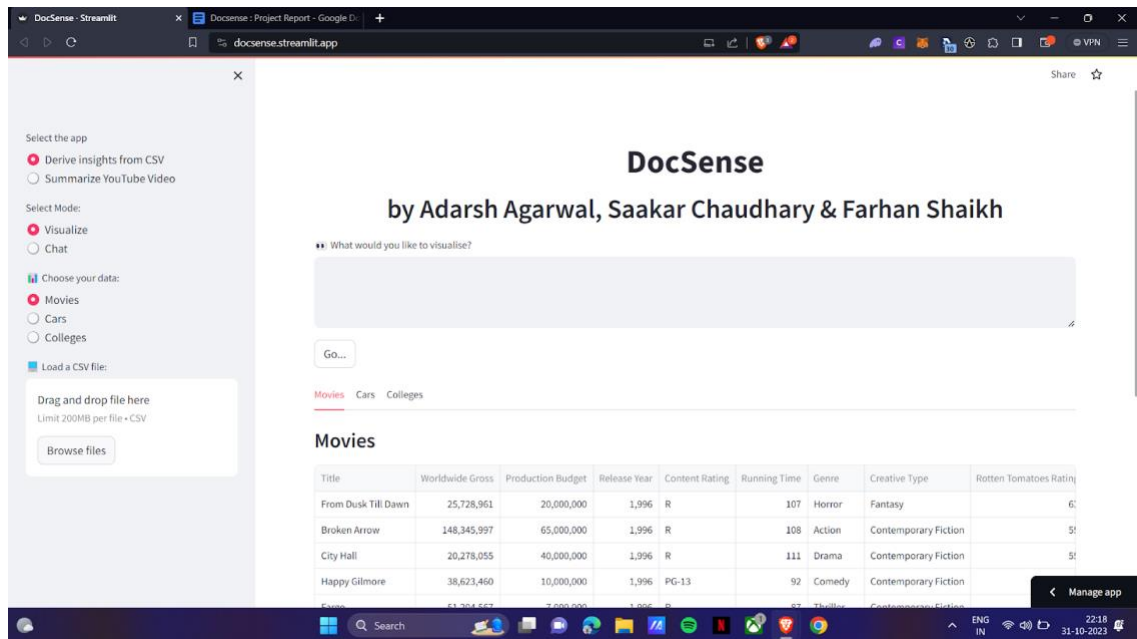


Fig 5 CSV Demonstration 1

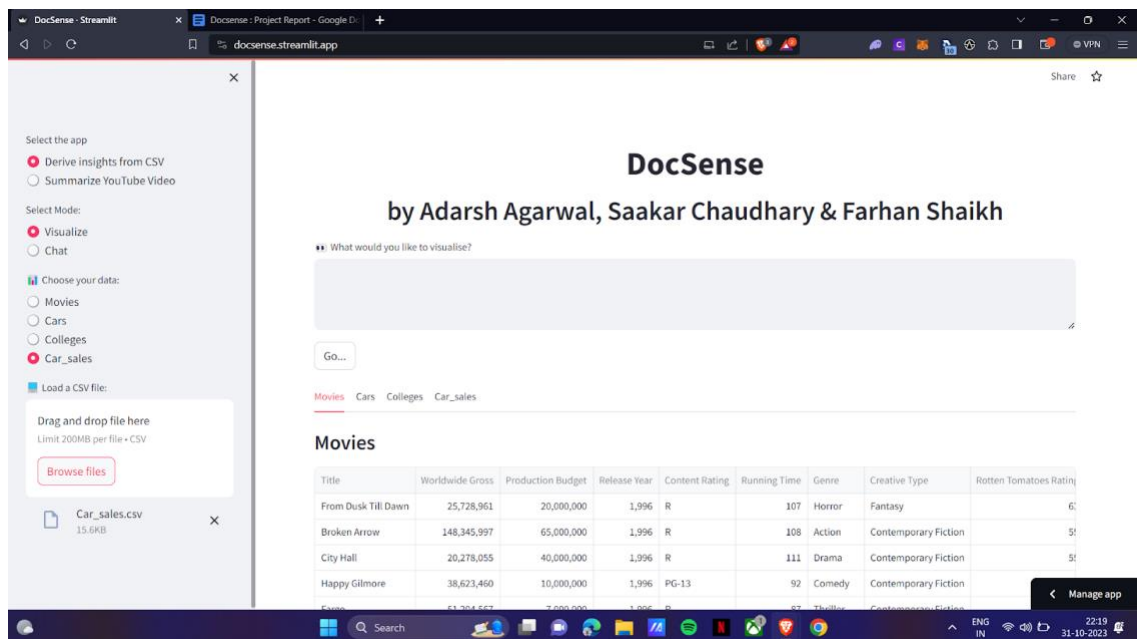


Fig 6 CSV Demonstration 2

DocSense

by Adarsh Agarwal, Saakar Chaudhary & Farhan Shaikh

What would you like to visualise?

Plot a Bar Graph for all the manufacturers based on collective sales in thousand

Go...

Fig 7 CSV Demonstration 3



Fig 8 CSV Demonstration 4

DocSense

by Adarsh Agarwal, Saakar Chaudhary & Farhan Shaikh

What would you like to visualise?

Plot a scatter plot for fuel efficiency vs horsepower

Go...

Fig 9 CSV Demonstration 5

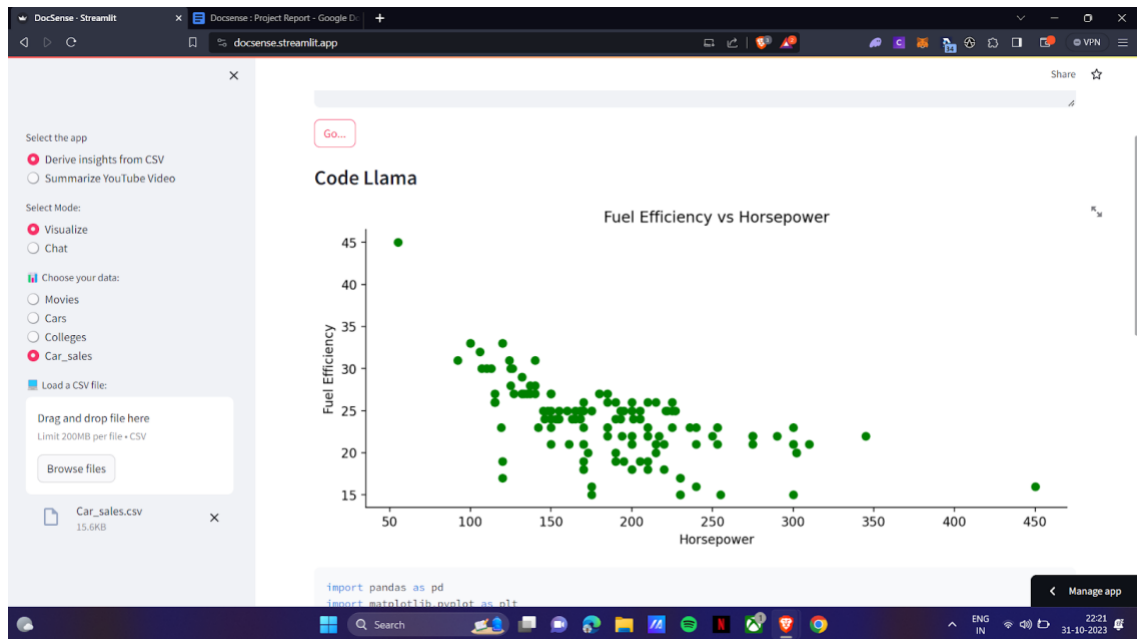


Fig 10 CSV Demonstration 6

DocSense

by Adarsh Agarwal, Saakar Chaudhary & Farhan Shaikh

What would you like to visualise?

Give a box plot for vehicle type and engine size

Go...

Fig 11 CSV Demonstration 7

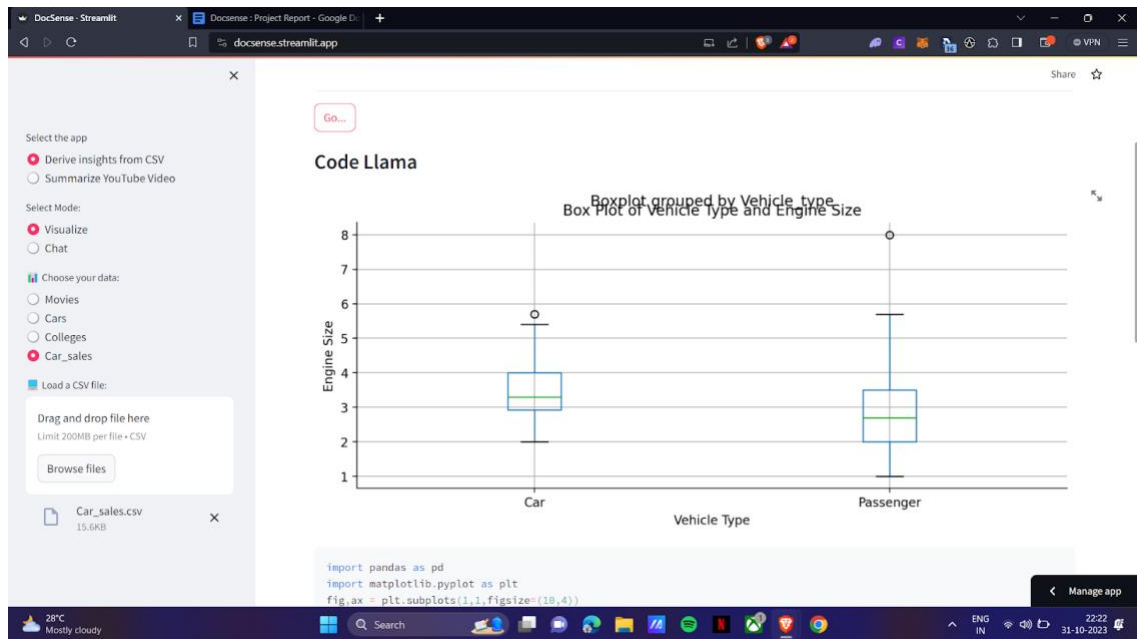


Fig 12 CSV Demonstration 8

Chat with CSV:

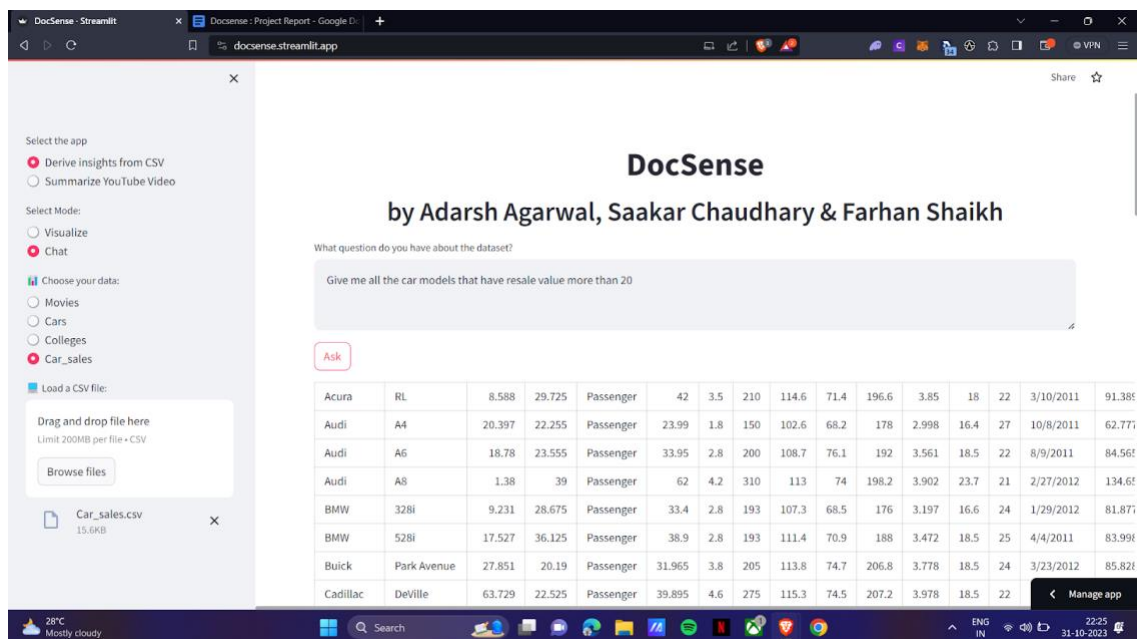


Fig 13 CSV Chat Demonstration 1

DocSense

by Adarsh Agarwal, Saakar Chaudhary & Farhan Shaikh

What question do you have about the dataset?

Give me all the car models that have resale value more than 20

Ask

Fig 14 CSV Chat Demonstration 2

DocSense - Streamlit

DocSense - Project Report - Google D...

docsense.streamlitapp/#by-adarsh-agarwal-saakar-chaudhary-farhan-shaikh

Select the app

- Derive insights from CSV
- Summarize YouTube Video

Select Mode:

- Visualize
- Chat

Choose your data:

- Movies
- Cars
- Colleges
- Car_sales

Load a CSV file:

Drag and drop file here
Limit 200MB per file • CSV

Browse files

Car_sales.csv
15.6KB

Brand	Model	Price	Mileage	Type	...
Porsche	Carrera Coupe	1.28	60.625	Passenger	...
Porsche	Carrera Cabrio	1.866	67.55	Passenger	...
Toyota	Land Cruiser	9.835	34.08	Car	...

```
import pandas as pd
from tabulate import tabulate
df=datasets["Car_sales"].copy()
df=df[df["_year_resale_value"]>20]
print(tabulate(df,tablefmt="html",showIndex=False))
```

Movies Cars Colleges Car_sales

Movies

Title	Worldwide Gross	Production Budget	Release Year	Content Rating	Running Time	Genre	Creative Type	Rotten Tomatoes Rating
From Dusk Till Dawn	25,728,961	20,000,000	1,996	R	107	Horror	Fantasy	6
Broken Arrow	148,345,997	65,000,000	1,996	R	108	Action	Contemporary Fiction	5
City Hall	20,278,055	40,000,000	1,996	R	111	Drama	Contemporary Fiction	5
Happy Gilmore	38,623,460	10,000,000	1,996	PG-13	92	Comedy	Contemporary Fiction	5
Fargo	51,204,567	7,000,000	1,996	R	87	Thriller	Contemporary Fiction	5
The Craft	55,669,466	15,000,000	1,996	R	100	Thriller	Fantasy	5

Manage app

Fig 15 CSV Chat Demonstration 3

DocSense

by Adarsh Agarwal, Saakar Chaudhary & Farhan Shaikh

What question do you have about the dataset?

Give me all the cars that have fuel capacity more than 15 and sales in thousand is more than 20

Ask

Fig 16 CSV Chat Demonstration 4

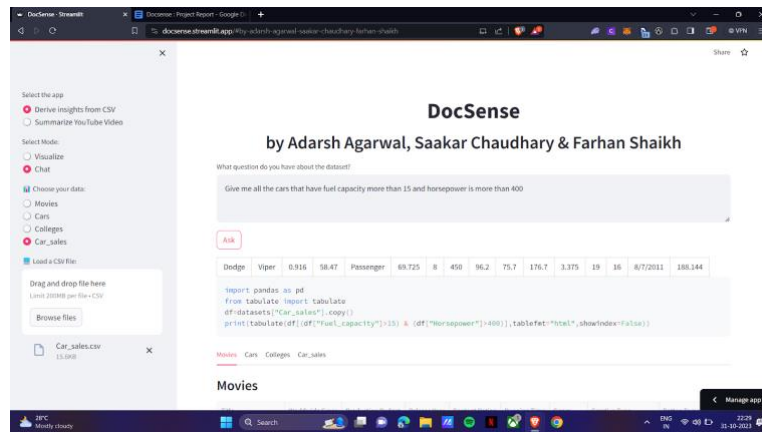


Fig 17 CSV Chat Demonstration 5

Youtube Summarization:

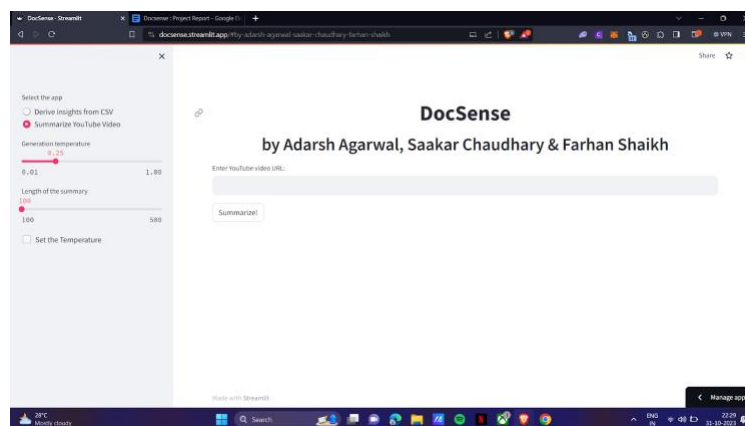


Fig 18 Youtube Summerizer Demonstration 1

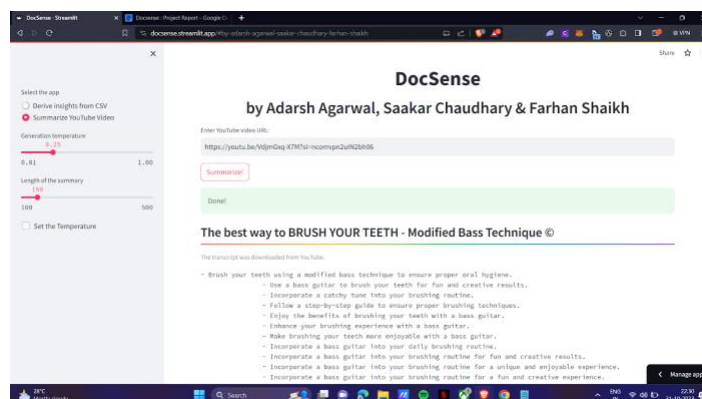


Fig 19 Youtube Summerizer Demonstration 2

4.4.2 Integration of External APIs

An further approach employed was the incorporation of external Application Programming Interfaces (APIs) within our system. The selection of these specialised APIs was made with the intention of transferring specific processing responsibilities from the local system, hence potentially enhancing response times. Preliminary examinations of external application programming interfaces (APIs) demonstrated encouraging outcomes in mitigating time lags for particular sorts of queries.

In order to achieve a successful integration of external APIs, it was necessary to conduct a comprehensive review and compatibility assessment to guarantee smooth operation within our system architecture.

4.4.3 Balancing Act: Accuracy vs. Responsiveness

The fundamental topic that emerged was the need to strike a fine balance between maintaining high response accuracy and improving response times. While prioritising the resolution of delays, it was of utmost importance to ensure the accuracy and validity of the provided responses. The aforementioned challenge underscored the necessity for thorough optimisations that involve not only the local processing capabilities but also the integration of external application programming interfaces (APIs).

4.4.4 Conclusion

In conclusion, our system consistently delivered accurate responses, showcasing the capabilities of our AI models. However, the challenge of delayed responses highlighted the importance of further optimization. Whether through hardware enhancements, external API integration, or striking the right balance between

accuracy and responsiveness, our commitment to delivering an exceptional user experience remains unwavering. Our ongoing efforts aim to ensure that users receive timely and precise answers while benefiting from the full potential of our AI-powered system.

Chapter 5 Advantages, Limitations and Applications

As with any innovative technology or solution, DocSense, the revolutionary AI-powered chatbot designed for document analysis and multimedia content, offers a range of benefits, has its own set of drawbacks, and operates within certain limitations. In this section, we will explore the advantages that make DocSense a game-changer, the disadvantages that may pose challenges, and the limitations that define the boundaries of its functionality. By understanding these aspects, users and developers can make informed decisions regarding the adoption and use of this powerful AI tool.

5.1 Advantages

DOCSense has several advantages when compared to regular chatbots including:

5.1.1 Interactive Data Visualization

One of DocSense's most potent features is its special ability to display CSV files in response to user inquiries. Through visually intuitive means, users can interactively examine and derive insights from their tabular data. This function improves data comprehension and decision-making by converting unstructured data into insightful graphs, charts, and other visual representations.

5.1.2 Data-Driven Decision Support

DocSense enables users to make data-driven decisions more successfully by providing real-time data visualisation. This feature offers a flexible and user-friendly method for finding patterns, trends, or anomalies in your CSV data.

5.1.3 Fast insights

A lot of the time, manual setup is necessary for traditional data visualisation tools. By using natural language queries to request particular visualisations

straight from DocSense, you may expedite the process of quickly gaining insights from your CSV data and save time.data.

5.1.4 User-Centric Flexibility

By allowing visualisations to be created in response to user inquiries, the visual representations are customised to meet the individual demands of the user. An experience with data analysis that is more efficient and customised is fostered by this user-centric approach.

5.1.5 Enhanced Data Exploration

Users are encouraged to conduct more thorough data explorations using DocSense's data visualisation capability. This benefit makes it easier to explore and analyse data at a deeper level, whether you're looking at sales numbers, financial statistics, or any other tabular information.

5.1.6 Cross-Domain Applicability

DocSense can be used in a variety of fields, such as business, research, education, and content creation, because of its adaptability. Because of its versatility, it can be used for a wide range of purposes.

5.1.7 Efficiency in terms of time and money

DocSense expedites the information extraction process from documents and multimedia content. For companies and professionals that depend on effective data analysis, this can result in considerable time and cost savings.

5.2 Disadvantages

DOCSense has some disadvantages which can result in some scenarios

5.2.1 Resource-Intensive Model

DocSense uses a 7 billion parameter Llama 2 language model to function. This model has the drawback of requiring a lot of resources, despite its exceptional

accuracy. Processing times may increase as a result, particularly when handling big data sets.

5.2.2 Restricted Parameter Size

Selecting a lesser model may incur various trade-offs as compared to the bigger Llama 2 model, which has 30 billion parameters. Even with its effective use of resources, the 7 billion parameter model might not be as accurate as the larger model for some jobs.

5.2.3 Performance vs. accuracy

Performance vs. accuracy is the trade-off that led to the adoption of the 7 billion parameter model. While it might yield answers more quickly, there's a chance that, in comparison to the larger model, the analysis's accuracy will occasionally be marginally degraded.

5.2.4 Processing Time

DocSense may have longer processing times depending on the task's complexity and the volume of data being analysed. This could be a drawback for users that need quick answers or insights.

5.2.5 Resource Requirement

The model's resource-intensive nature may need substantial computational resources, which could restrict its usability for customers with low processing capacity or those accessing the platform through mobile devices.

5.2.6 Cost of Scaling

Scaling the infrastructure to maintain acceptable processing speeds may need more expenditures as the user base and data volume increase, providing a financial hurdle for enterprises wishing to implement DocSense at scale.

5.3 Limitations

Here are some limitations we have identified with DOCSense

5.3.1 Language Model Scope

The scope and knowledge incorporated in the underlying Llama 2 language model serve as boundaries for DocSense's natural language comprehension and summarising skills. Beyond its training data, it might not be aware of very recent advancements, specialised subjects, or particular industry lingo, which could restrict its capacity to offer insights in some areas.

5.3.2 Variations in Accuracy

Although DocSense aims for precise document summary, there are some variations in accuracy. Depending on the intricacy and content of the input documents, the quality of the summaries that are produced may differ, sometimes resulting in errors or omissions.

5.3.3 Limitations with Multimedia Content

DocSense is capable of analysing YouTube videos, however its content analysis is based on speech-to-text transcription. This means that factors like background noise, accents, and the quality of the audio in films could affect how well it performs, as could the existence of non-verbal elements like graphics or images.

5.3.4 Concerns about Privacy and Security

Adding private or confidential documents to DocSense may give rise to privacy and security issues. When sharing such material, users should proceed with caution. Organisations should consider the consequences of outsourcing document analysis to outside providers.

5.3.5 Resource and Scalability Constraints

When working with a large number of concurrent users or particularly big data sets, the resource-intensive nature of the underlying AI model may cause

scalability issues. Expanding the infrastructure to accommodate the increasing demand may provide both financial and technological challenges.

5.3.6 Latency

When producing results, processing documents, videos, and intricate inquiries might cause delays. Applications that require instantaneous answers in real-time might not be appropriate for this.

5.3.7 User Skill Level

Although DocSense is meant to be easy to use, people with little experience with data analysis or technology may find it difficult to fully utilise its potential. It might need sufficient training and assistance for users to realise the full potential of it.

Chapter 6 Conclusion and Future Scope

We have discovered DocSense's remarkable advantages, identified some possible disadvantages, and comprehended the limitations of this cutting-edge AI-powered chatbot for document analysis and multimedia content. It's time to evaluate the value of DocSense's current capabilities as we draw to a close and, more importantly, think about the fascinating opportunities that its future development holds.

In this last section, we will summarise the key findings from our analysis of DocSense's benefits, drawbacks, and constraints. We will also look ahead and consider future developments that could strengthen DocSense's position as a revolutionary tool in the digital age. By doing this, we can recognise the progress that has been done and look forward to the promising directions that an AI chatbot may go in order to better meet the always changing needs of its users and the larger context of knowledge dissemination.

6.1 Conclusion

The introduction of DocSense, the AI-powered chatbot for document analysis and multimedia material, marks a significant breakthrough in the field of data processing and knowledge extraction in the era of rapidly expanding information and digital transformation. Upon contemplating the trip, its potential, benefits, drawbacks, and restrictions, a few important conclusions emerge.

DocSense's many advantages—such as its precise summarization, extensive document processing, and natural language interaction—make it an adaptable tool that has the power to revolutionise the way we extract insights from data. Its special capacity to evaluate YouTube videos and provide data visualisation via user queries provides features that support a wide variety of users, from decision-makers to content creators.

But every innovation has its limitations. DocSense creates trade-offs between accuracy and performance by relying on a resource-intensive 7 billion parameter AI model instead of a more expansive 30 billion parameter version. It is important to take into account processing time, resource needs, and potential scalability issues, particularly in situations with high demand.

DocSense has a bright future ahead of it, despite these drawbacks. Improvements can be made by ongoing model upgrades, processing speed optimisation, and possible integrations with larger datasets. Its ability to increase precision while preserving efficiency is still an appealing possibility.

In summary, DocSense stands out as a ground-breaking technology that bridges the divide between conventional document analysis and the constantly growing volume of digital data. Its benefits provide an indication of the revolutionary potential of AI in information extraction. The path does not end here, even while the current limits highlight the necessity for careful deliberation in its application. The future roadmap calls for ongoing improvement with the goal of fully using AI's potential to assist researchers, decision-makers, content producers, and knowledge disseminators in a variety of fields. With an eye towards the future, DocSense pledges to change and grow to satisfy the changing demands of a society that is becoming more and more focused on data.

6.2 Future Scope

The path ahead for DocSense is exciting and full of opportunities to significantly advance its skills and adaptability in the ever-changing field of knowledge extraction and data analysis. One of the most exciting future developments is the addition of generic chatbot features, which would allow DocSense to converse with people in deeper, more involved ways. With this update, users will be able to engage in two-way discussion where they may ask questions, discuss, and learn. This is expected to greatly increase user engagement.

DocSense's ability to use URLs to summarise articles provides an intriguing path for future development. With the help of this function, users may quickly gain insights into online information and save time by just entering the URL of a web article. DocSense quickly generates a succinct description. This improvement is especially important in a time when the internet is a huge information source and it is essential to be able to extract useful information from online articles.

DocSense is ready to expand into multimedia in order to enhance its capabilities beyond text-based information. This calls for in-depth image and audio material analysis and interpretation. DocSense will be able to deliver insights from both visual and audible sources thanks to the integration of computer vision and audio recognition technologies, opening up a whole new realm of data analysis. Marketing professionals, researchers, and content creators who work with various media platforms may find this development particularly interesting.

One of the main goals of DocSense's future growth is language diversity. Although it speaks English and other major languages fluently now, the goal is to increase its multilingual support. With this extension, DocSense would be able to process and interpret a wider variety of languages and dialects, thereby serving a global audience. This diversity is in line with the need for multilingual data analysis tools and the world's growing interconnectedness.

Another important factor for the future to take into account is scalability. Scalability improvements are critical as user demands rise and organisations look to employ DocSense for larger datasets. These improvements will guarantee that DocSense stays a dependable and effective tool that can manage a growing workload without sacrificing functionality.

DocSense is dedicated to enhancing its features to protect sensitive data in an era where data security and privacy are critical. Users can feel secure knowing that their data is protected thanks to the integration of encryption, access controls, and strong data protection mechanisms.

One more aspect of progress in the future is personalization. Giving users the option to personalise every aspect of their DocSense experience—from visualisation styles to preferred summarising methods—guarantees that the tool will satisfy different needs and tastes. This flexibility gives users more control and customisation.

Last but not least, a big step towards offering a thorough and smooth data analysis environment is DocSense's capability for integration with other data analysis tools. Integration with commonly used tools can improve users' overall analytical experience by streamlining data workflows.

To sum up, DocSense will continue to innovate, adapt, and improve user experience along the way. With its ability to accommodate many content kinds, languages, and communication idioms, it is positioned to influence how data analysis, information sharing, and dynamic interactions are conducted in the digital era. DocSense is headed towards becoming an essential tool for professionals in a variety of fields, thanks to its persistent dedication to satisfying the changing demands of both users and the larger data-driven world. With open arms, DocSense is prepared to welcome the bright future and push the boundaries of AI-driven knowledge distribution.

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Appendix A: Code

Streamlit_app.py

```

import pandas as pd
import time
import streamlit as st
import warnings
import os
import warnings
import sys
from io import StringIO
import contextlib
from tabulate import tabulate
import streamlit as st
import pytube
from pytube import YouTube
import transformers
import torch
from transformers import AutoTokenizer, AutoModelForCausalLM
from youtube_transcript_api import YouTubeTranscriptApi
from huggingface_hub import InferenceClient
import librosa
import soundfile as sf
from langchain.chains.llm import LLMChain
from langchain.prompts import PromptTemplate
from langchain.chains import ReduceDocumentsChain, MapReduceDocumentsChain
from langchain.chains.combine_documents.stuff import StuffDocumentsChain
from langchain import HuggingFaceHub
import gradio as gr
from classes import get_primer, format_question, run_request, get_text_primer

warnings.filterwarnings("ignore")
st.set_option('deprecation.showPyplotGlobalUse', False)
st.set_page_config(layout="wide", page_title="DocSense")

st.markdown("<h1 style='text-align: center; font-weight:bold; font-family:'Trebuchet MS', sans-serif; padding-top: 0rem;'>DocSense</h1>",
, unsafe_allow_html=True)
st.markdown("<h2 style='text-align: center; padding-top: 0rem;'>by Adarsh Agarwal, Saakar Chaudhary & Farhan Shaikh</h2>",
unsafe_allow_html=True)
hf_key = 'hf_WrpvXrHBakDJyTBBY0XezwPipImSmsMKAK'

placeholder = st.empty()

app_selection = st.sidebar.radio
("Select the app", ["Derive insights from CSV", "Summarize YouTube Video"])

if app_selection == "Derive insights from CSV":
    placeholder.empty()
    if "datasets" not in st.session_state:
        datasets = {}
        datasets["Movies"] = pd.read_csv("movies.csv")
        datasets["Cars"] = pd.read_csv("cars.csv")
        datasets["Colleges"] = pd.read_csv("colleges.csv")
        st.session_state["datasets"] = datasets
    else:
        datasets = st.session_state["datasets"]

    with placeholder.container():
        with st.sidebar:
            mode = st.radio("Select Mode:", ("Visualize", "Chat"))

            dataset_container = st.empty()

            try:
                uploaded_file = st.file_uploader
                (":computer: Load a CSV file:", type="csv")
                index_no=0
                if uploaded_file:

```

```

file_name = uploaded_file.name[:-4].capitalize()
    datasets[file_name] = pd.read_csv(uploaded_file)
    index_no = len(datasets)-1
except Exception as e:
    st.error("File failed to load. Please select a valid CSV file.")
    print("File failed to load.\n" + str(e))

chosen_dataset = dataset_container.radio
(":bar_chart: Choose your data:",
 datasets.keys(),index=index_no)#,horizontal=True,)

if mode == "Visualize":
    question = st.text_area
    (":eyes: What would you like to visualise?", height=10)
    go_btn = st.button("Go...")

    if go_btn:
        if not hf_key.startswith('hf_'):
            st.error("Please enter a valid HuggingFace API key.")
        else:
            st.subheader("Code Llama")
            try:

                primer1, primer2 =
                get_primer(datasets[chosen_dataset], 'datasets["'
                + chosen_dataset + "']')

                question_to_ask =
                format_question(primer1, primer2, question, "Code Llama")

                answer = run_request
                (question_to_ask, "CodeLlama-34b-Instruct-hf"
                , alt_key=hf_key)

                answer = primer2 + answer
                st.code(answer)
                plot_area = st.empty()
                plot_area.pyplot(exec(answer))
            except Exception as e:
                st.error(f"An error occurred: {e}")

elif mode == "Chat":
    chat_question = st.text_area
    ("What question do you have about the dataset?", height=10)
    chat_btn = st.button("Ask")

    @contextlib.contextmanager
    def stdoutIO(stdout=None):
        old = sys.stdout
        if stdout is None:
            stdout = StringIO()
        sys.stdout = stdout
        yield stdout
        sys.stdout = old

    if chat_btn:
        try:
            output_dict={}
            primer1, primer2 = get_text_primer
            (datasets[chosen_dataset],
            'datasets["' + chosen_dataset + "']')
            question_to_ask = format_question
            (primer1, primer2, chat_question, "Code Llama")
            answer = run_request
            (question_to_ask,
            "CodeLlama-34b-Instruct-hf", alt_key=hf_key)
            answer = primer2 + answer
            output_dict['datasets'] = datasets
            op_area = st.empty()

```

```

        with op_area.container():
            st.code(answer)
            with stdoutIO() as s:
                exec(answer)
            st.markdown
            (s.getvalue()),
        except Exception as e:
            st.error(f"An error occurred: {e}")

    tab_list = st.tabs(datasets.keys())

    for dataset_num, tab in enumerate(tab_list):
        with tab:

            dataset_name = list(datasets.keys())[dataset_num]
            st.subheader(dataset_name)
            st.dataframe(datasets[dataset_name], hide_index=True)

    footer="""<style>.footer {position: fixed;left: 0;bottom:
        0;width: 100%;text-align: center;}</style><div class="footer">
        <p> <a style='display: block; text-align: center;'>
        Datasets courtesy of NL4DV, nvBench and ADVISor </a></p></div>"""
    st.caption("Datasets courtesy of NL4DV, nvBench and ADVISor")

    hide_streamlit_style = """
        <style>
        #MainMenu {visibility: hidden;}
        footer {visibility: hidden;}
        </style>
        """
    st.markdown(hide_streamlit_style, unsafe_allow_html=True)

elif app_selection == "Summarize YouTube Video":
    placeholder.empty()
    transcription_model_id = "openai/whisper-large"
    llm_model_id = "tiiuae/falcon-7b-instruct"
    HF_TOKEN = os.environ.get("HF_TOKEN", None)
    save_dir = "docs"

    def get_youtube_title(url):
        yt = YouTube(url)
        return yt.title

    def summarize_youtube_video(url,
        force_transcribe, hf_key, temperature, words, do_sample):
        save_dir = os.path.join('./', 'docs')
        if not os.path.exists(save_dir):
            os.mkdir(save_dir)

        title, text, transcript_source =
            transcribe_youtube_video(url, force_transcribe, True, hf_key)

        summary, summary_source =
            summarize_text(title, text, temperature, words, True, hf_key, do_sample)

        return title, text, transcript_source, summary, summary_source

    def transcribe_youtube_video(url, force_transcribe, use_api, hf_key):
        yt = YouTube(str(url))
        text = ''

        try:
            text = get_yt_transcript(url)
        except:
            pass

```

```

if text == '' or force_transcribe:
    if use_api:
        text = transcribe_yt_vid_api(url, hf_key=hf_key)
        transcript_source =
        'The transcript was generated using {} via the Hugging Face Hub
API.'
        .format(transcription_model_id)
    else:
        text = transcribe_yt_vid(url)
        transcript_source =
        'The transcript was generated using {} hosted locally.'
        .format(transcription_model_id)
    else:
        transcript_source = 'The transcript was downloaded from YouTube.'

    return yt.title, text, transcript_source

def get_yt_transcript(url):
    text = ''
    vid_id = pytube.extract.video_id(url)
    temp = YouTubeTranscriptApi.get_transcript(vid_id)
    for t in temp:
        text += t['text'] + ' '
    return text

def transcribe_yt_vid_api(url, hf_key):
    yt = YouTube(str(url))
    audio = yt.streams.filter(only_audio=True).first()
    out_file = audio.download(filename="audio.wav", output_path=save_dir)

    text = ''
    t = 25
    x, sr = librosa.load(out_file, sr=None)

    for i in range(0, (len(x) // (t * sr)) + 1):
        y = x[t * sr * i: t * sr * (i + 1)]
        split_path = os.path.join(save_dir, "audio_split.wav")
        sf.write(split_path, y, sr)
        text += client.automatic_speech_recognition(split_path)

    return text

def summarize_text(title, text, temperature, words, use_api, hf_key, do_sample):

    from langchain.chains.llm import LLMChain
    from langchain.prompts import PromptTemplate
    from langchain.chains import ReduceDocumentsChain, MapReduceDocumentsChain
    from langchain.chains.combine_documents.stuff import StuffDocumentsChain
    import torch
    import transformers
    from transformers import BitsAndBytesConfig
    from transformers import AutoTokenizer, AutoModelForCausalLM

    from langchain import HuggingFacePipeline
    import torch

    model_kwargs1 = {"temperature":temperature ,
                     "do_sample":do_sample,
                     "min_new_tokens":200-25,
                     "max_new_tokens":200+25
                    }
    model_kwargs2 = {"temperature":temperature ,
                     "do_sample":do_sample,
                     "min_new_tokens":words-25,
                     "max_new_tokens":words+25,
                     'repetition_penalty':2.0
                    }

```

```

    }

    if not do_sample:
        del model_kwargs1["temperature"]
        del model_kwargs2["temperature"]

    if use_api:

        llm=HuggingFaceHub(
            repo_id=llm_model_id, model_kwargs=model_kwargs1,
            huggingfacehub_api_token=hf_key
        )
        llm2=HuggingFaceHub(
            repo_id=llm_model_id, model_kwargs=model_kwargs2,
            huggingfacehub_api_token=hf_key
        )
        summary_source =
        'The summary was generated using {} via Hugging Face API.'
        .format(llm_model_id)

    else:
        quantization_config = BitsAndBytesConfig(
            load_in_4bit=True,
            bnb_4bit_compute_dtype=torch.float16,
            bnb_4bit_quant_type="nf4",
            bnb_4bit_use_double_quant=True,
        )

        tokenizer = AutoTokenizer.from_pretrained(llm_model_id)
        model = AutoModelForCausalLM.from_pretrained(llm_model_id,
                                                    # quantization_config=quantization_config
                                                    )

        model.to_bettertransformer()

        pipeline = transformers.pipeline(
            "text-generation",
            model=model,
            tokenizer=tokenizer,
            torch_dtype=torch.bfloat16,
            device_map="auto",
            pad_token_id=tokenizer.eos_token_id,
            **model_kwargs1,
        )
        pipeline2 = transformers.pipeline(
            "text-generation",
            model=model,
            tokenizer=tokenizer,
            torch_dtype=torch.bfloat16,
            device_map="auto",
            pad_token_id=tokenizer.eos_token_id,
            **model_kwargs2,
        )
        llm = HuggingFacePipeline(pipeline=pipeline)
        llm2 = HuggingFacePipeline(pipeline=pipeline2)

        summary_source =
        'The summary was generated using {} hosted
        locally.'.format(llm_model_id)

    # Map
    map_template = """
    You are an intelligent AI assistant that is tasked
    to review the content of a video and provide a concise and accurate summary.
    \n
    You do not provide information that is not mentioned in the video.
    You only provide information that you are absolutely sure about.\n
    Summarize the following text in a clear and concise way:\n
    ----- \n

```

```

TITLE: `{title}`\n
TEXT:\n
`{docs}`\n
----- \n
BRIEF SUMMARY:\n
"""
map_prompt = PromptTemplate(
    template = map_template,
    input_variables = ['title','docs']
)
map_chain = LLMChain(llm=llm, prompt=map_prompt)

# Reduce - Collapse
collapse_template = """
You are an intelligent AI assistant that is tasked
to review the content of a video and provide a concise and accurate
summary.\n
You do not provide information that is not mentioned in the video.
You only provide information that you are absolutely sure about.\n
The following is set of partial summaries of a video:\n
----- \n
TITLE: `{title}`\n
PARTIAL SUMMARIES:\n
`{doc_summaries}`\n
----- \n
Take these and distill them into a consolidated summary.\n
SUMMARY:\n
"""

collapse_prompt = PromptTemplate(
    template = collapse_template,
    input_variables = ['title','doc_summaries']
)
collapse_chain = LLMChain(llm=llm, prompt=collapse_prompt)

collapse_documents_chain = StuffDocumentsChain(
    llm_chain=collapse_chain, document_variable_name="doc_summaries"
)

# Final Reduce - Combine
combine_template = """\n
You are an intelligent AI assistant that is tasked to review the content of
a video and provide a concise and accurate summary.\n
You do not provide information that is not mentioned in the video.
You only provide information that you are absolutely sure about.\n
The following is a set of partial summaries of a video:\n
----- \n
TITLE: `{title}`\n
PARTIAL SUMMARIES:\n
`{doc_summaries}`\n
----- \n
Generate an executive summary of the whole text in maximum {words}
words that contains the main messages, points, and arguments presented
in the video as bullet points.\n
EXECUTIVE SUMMARY:\n
"""

combine_prompt = PromptTemplate(
    template = combine_template,
    input_variables = ['title','doc_summaries','words']
)
combine_chain = LLMChain(llm=llm2, prompt=combine_prompt)

combine_documents_chain = StuffDocumentsChain(
    llm_chain=combine_chain, document_variable_name="doc_summaries"
)

```

```

reduce_documents_chain = ReduceDocumentsChain(
    # This is final chain that is called.
    combine_documents_chain=combine_documents_chain,
    # If documents exceed context for `StuffDocumentsChain`
    collapse_documents_chain=collapse_documents_chain,
    # The maximum number of tokens to group documents into.
    token_max=800,
)

map_reduce_chain = MapReduceDocumentsChain(
    # Map chain
    llm_chain=map_chain,
    # Reduce chain
    reduce_documents_chain=reduce_documents_chain,
    # The variable name in the llm_chain to put the documents in
    document_variable_name="docs",
    # Return the results of the map steps in the output
    return_intermediate_steps=False,
)

from langchain.document_loaders import TextLoader
from langchain.text_splitter import TokenTextSplitter

if not os.path.exists(save_dir):
    os.makedirs(save_dir)

transcript_file_path = os.path.join(save_dir, 'transcript.txt')
with open(save_dir+'transcript.txt','w') as f:
    f.write(text)
loader = TextLoader(save_dir+"transcript.txt")
doc = loader.load()
text_splitter = TokenTextSplitter(chunk_size=800, chunk_overlap=100)
docs = text_splitter.split_documents(doc)

summary = map_reduce_chain.run({'input_documents':docs, 'title':title, 'words':words})

try:
    del(map_reduce_chain,reduce_documents_chain,combine_chain,
collapse_documents_chain,map_chain,collapse_chain,llm,llm2,pipeline,pipeline2,model,tokenizer)
except:
    pass
torch.cuda.empty_cache()

return summary, summary_source

import gradio as gr
import pytube
from pytube import YouTube
force_transcribe = False
temperature = st.sidebar.slider("Generation temperature", min_value=0.01, max_value=1.0,
value=0.25)
words = st.sidebar.slider("Length of the summary", min_value=100, max_value=500, value=100)
do_sample = st.sidebar.checkbox("Set the Temperature")

with placeholder.container():
    url = st.text_input("Enter YouTube video URL:", "")
    if st.button("Summarize!"):
        if not url:
            st.error("Please enter a valid YouTube URL.")
        else:
            with st.spinner('Processing, please wait....'):
                time.sleep(7)
            st.success('Done!')
            title, text, transcript_source = transcribe_youtube_video(url,
force_transcribe, True, hf_key)
            summary, summary_source = summarize_text(title, text, temperature,
words, True, hf_key, do_sample)

            st.subheader(title, divider='rainbow')
            st.caption(transcript_source)
            st.text(summary)

```

Classes.py

```

from langchain import HuggingFaceHub, LLMChain, PromptTemplate

def run_request(question_to_ask, model_type, alt_key):
    # Hugging Face model
    llm = HuggingFaceHub(huggingfacehub_api_token = alt_key, repo_id="codellama/" + model_type,
    model_kwargs={"temperature":0.1, "max_new_tokens":500})
    llm_prompt = PromptTemplate.from_template(question_to_ask)
    llm_chain = LLMChain(llm=llm, prompt=llm_prompt)
    llm_response = llm_chain.predict()
    # rejig the response
    llm_response = format_response(llm_response)
    return llm_response

def format_response(res):
    # Remove the load_csv from the answer if it exists
    csv_line = res.find("read_csv")
    if csv_line > 0:
        return_before_csv_line = res[0:csv_line].rfind("\n")
        if return_before_csv_line == -1:
            # The read_csv line is the first line so there is nothing to need before it
            res_before = ""
        else:
            res_before = res[0:return_before_csv_line]
        res_after = res[csv_line:]
        return_after_csv_line = res_after.find("\n")
        if return_after_csv_line == -1:
            # The read_csv is the last line
            res_after = ""
        else:
            res_after = res_after[return_after_csv_line:]
        res = res_before + res_after
    return res

def format_question(primer_desc, primer_code, question, model_type):
    # Fill in the model_specific_instructions variable
    instructions = "\nDo not use the 'c' argument in the plot function, use 'color' instead and only
    pass color names like 'green', 'red', 'blue'."
    primer_desc = primer_desc.format(instructions)
    # Put the question at the end of the description primer within quotes, then add on the code primer.
    return ""'\n' + primer_desc + question + '\n""'\n' + primer_code

def get_primer(df_dataset, df_name):
    # Primer function to take a dataframe and its name
    # and the name of the columns
    # and any columns with less than 20 unique values it adds the values to the primer
    # and horizontal grid lines and labeling
    primer_desc = "Use a dataframe called df from data_file.csv with columns '" \
    + "', '".join(str(x) for x in df_dataset.columns) + "'. "
    for i in df_dataset.columns:
        if len(df_dataset[i].drop_duplicates()) < 20 and df_dataset.dtypes[i]=="0":
            primer_desc = primer_desc + "\nThe column '" + i + "' has categorical values '" + \
            "', '".join(str(x) for x in df_dataset[i].drop_duplicates()) + "'. "
        elif df_dataset.dtypes[i]=="int64" or df_dataset.dtypes[i]=="float64":
            primer_desc = primer_desc + "\nThe column '" + i + "' is type " + str(df_dataset.dtypes[i])
    + " and contains numeric values. "
    primer_desc = primer_desc + "\nLabel the x and y axes appropriately."
    primer_desc = primer_desc + "\nAdd a title. Don't put any fig supptitle."
    primer_desc = primer_desc + "{}" # Space for additional instructions if needed
    primer_desc = primer_desc + "\nUsing Python version 3.9.12, create a script using the dataframe df
    to graph the following: "
    primer_code = "import pandas as pd\nimport matplotlib.pyplot as plt\n"
    primer_code = primer_code + "fig,ax = plt.subplots(1,1,figsize=(10,4))\n"
    primer_code = primer_code +
    "ax.spines['top'].set_visible(False)\nax.spines['right'].set_visible(False) \n"
    primer_code = primer_code + "df=" + df_name + ".copy()\n"
    return primer_desc, primer_code

def get_text_primer(df_dataset, df_name):
    # Primer function to take a dataframe
    # and the name of the columns
    # and any columns with less than 20 unique values it adds the values to the primer
    primer_desc = "Use a dataframe called df from data_file.csv with columns '" \
    + "', '".join(str(x) for x in df_dataset.columns) + "'. "
    for i in df_dataset.columns:
        if len(df_dataset[i].drop_duplicates()) < 20 and df_dataset.dtypes[i]=="0":
            primer_desc = primer_desc + "\nThe column '" + i + "' has categorical values '" + \
            "', '".join(str(x) for x in df_dataset[i].drop_duplicates()) + "'. "
        elif df_dataset.dtypes[i]=="int64" or df_dataset.dtypes[i]=="float64":
            primer_desc = primer_desc + "\nThe column '" + i + "' is of type " +
    str(df_dataset.dtypes[i]) + " and contains numeric values. "
    primer_desc = primer_desc + "\nPlease ensure that the output is generated using 'print' statement
    with tabulate only, keep tablefmt as html and showindex=False.\n Make sure to show headers of the
    dataframe."
    primer_desc = primer_desc + "\nUsing Python version 3.9.12, create a script using the dataframe df
    to answer the following:"
    primer_code = "import pandas as pd \nfrom tabulate import tabulate \n"
    primer_code = primer_code + "df=" + df_name + ".copy()\n"
    return primer_desc, primer_code

```